



22-MAVZU · III BOB. PYTHON VA MA'LUMOTLAR BAZASI

# Python control flow: shartlar va sikllar. Fayllar bilan ishlash

if, elif, else va chekinish qoidasi; match; for va range(), while, break, continue; try/except bilan xatolarni ushlash; fayllarni ochish, o'qish, yozish va qo'shish (with open). Python kalkulyatori.

# Bugun nimani o'rganamiz?

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- Shartlar: `if`, `elif`, `else` va **chekinish** qoidasi
- Sikllar: `for` + `range()`, `while`, `break`, `continue`
- Xatolarni ushlash: `try` / `except`
- **Fayllar:** yaratish, yozish, o'qish, qo'shish
- **Python kalkulyatori** (15-loyiha)

# if, elif, else

Python

```
ball = int(input("Ballingiz: "))

if ball ≥ 86:
    print("A'lo! 🏆")
elif ball ≥ 71:
    print("Yaxshi 👍")
elif ball ≥ 56:
    print("Qoniqarli")
else:
    print("Qoniqarsiz 😞")
```

*# JS'dagi "else if"*

- Shart oxirida **ikki nuqta** :
- **else if** emas — **elif**

# Chekinish — Python'ning eng muhim qoidasi

Python

```
yosh = 15
if yosh ≥ 14:
    print("Pasport olish mumkin")      # ← if ichida (4 bo'sh joy)
    print("Tabriklaymiz!")             # ← if ichida
print("Bu har doim chiqadi")          # ← tashqarida
```

- JS'da bloklar `{ }` bilan, Python'da — `:` va **4 bo'sh joy**



**Kitob mundarijasi:** kichik bo'limlar o'ngroqqa surilgan — qaysi bo'limga tegishli ekanini siljishdan bilasiz.



**IndentationError** — chekinish noto'g'ri. Bo'sh joy va Tab'ni aralashtirmang

# Qo'shimcha imkoniyatlar

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Python

```
holat = "katta" if yosh ≥ 18 else "voyaga yetmagan"    # ternary

if 12 ≤ yosh ≤ 18:                                     # oraliq – JS'da bunday yozib bo'lmadi!
    print("Maktab yoshi")

match kun:                                              # Python 3.10+ – switch o'rniga
    case 6 | 7:
        print("Dam olish 🏖️")
    case _:
        print("Ish kuni")
```

# for va range()

Python

```
for i in range(5):           # 0 1 2 3 4    (5 kirmaydi!)
    print(i, end=" ")
for i in range(1, 6):       # 1 2 3 4 5
    ...
for i in range(10, 0, -2):   # 10 8 6 4 2
    ...
for harf in "Python":       # har bir harf
    print(harf, end="-")
```

JavaScript

```
for (let i = 0; i < 5; i++)
```

```
for (let i = 1; i ≤ 10; i += 2)
```

```
for (const x of arr)
```

Python

```
for i in range(5):
```

```
for i in range(1, 11, 2):
```

```
for x in arr:
```

# Yig'indi va ko'paytirish jadvali

Python

```
jami = 0
for i in range(1, 101):
    jami += i
print("1..100 yig'indi:", jami)    # 5050

son = int(input("Qaysi sonning jadvali? "))
for i in range(1, 11):
    print(f"{son} × {i:2} = {son * i:3}")    # :2, :3 – tekislash
```

# while, break, continue

Python

```
import random
yashirin = random.randint(1, 20)
urinish = 0

while True:                                # break bilan chiqamiz
    taxmin = int(input("1-20 oraliq'ida son: "))
    urinish += 1
    if taxmin == yashirin:
        print(f"🎉 Topdingiz! {urinish} ta urinishda")
        break
    elif taxmin < yashirin:
        print("⬆ Kattaroq")
    else:
        print("⬇ Kichikroq")
```

- `continue` — keyingi qadamga o'tish

# try / except — xatolarni ushlash

Python

```
try:
    son = int(input("Son kiriting: "))
    print("Natija:", 100 / son)
except ValueError:
    print("❌ Bu son emas!")
except ZeroDivisionError:
    print("❌ Nolga bo'lib bo'lmaydi!")
finally:
    print("Dastur tugadi")          # har doim bajariladi
```

Natija

```
Son kiriting: abc
❌ Bu son emas!
Dastur tugadi
```

- `int("salom")` dasturni to'xtatadi — `try` uni "yiqilish"dan saqlaydi

BO'LIM

# Fayllar bilan ishlash

Dastur yopilsa ham ma'lumot saqlansin

# Faylga yozish

Python

```
with open("eslatma.txt", "w", encoding="utf-8") as f:
    f.write("Birinch qator\n")    # \n — yangi qator
    f.write("Ikkinchi qator\n")
```

- `with ... as f:` — ish tugagach faylni **avtomatik yopadi**
- `encoding="utf-8"` — o', g' va emoji uchun. **Doim yozing!**

| Rejim | Ma'nosi           | Fayl bor bo'lsa                   |
|-------|-------------------|-----------------------------------|
| "r"   | O'qish (standart) | O'qiydi (yo'q bo'lsa — xato)      |
| "w"   | Yozish            | ⚠ Eski mazmunni <b>o'chiradi!</b> |
| "a"   | Qo'shish          | Oxiriga qo'shadi                  |

# Fayldan o'qish va qo'shish

Python

```
with open("eslatma.txt", encoding="utf-8") as f:
    matn = f.read()                # butun fayl

with open("eslatma.txt", encoding="utf-8") as f:
    for i, qator in enumerate(f, start=1): # qatorma-qator
        print(i, qator.strip())          # strip() - \n ni olib tashlaydi

import os
with open("kundalik.txt", "a", encoding="utf-8") as f:
    f.write(input("Bugun nima qildingiz? ") + "\n")
if os.path.exists("kundalik.txt"):
    ...
```

# Amaliyot: Python kalkulyatori (15-loyiha)

Python

```
def son_sora(matn):
    while True:
        try:
            return float(input(matn))
        except ValueError:
            print(" ⚠ Iltimos, son kiriting")

while True:
    amal = input("Amal (+ - * / ** yoki 'chiqish', 'tarix'): ")
    if amal == "chiqish":
        break
    if amal == "tarix":
        with open("tarix.txt", encoding="utf-8") as f:
            print(f.read())
        continue
    a, b = son_sora("1-son: "), son_sora("2-son: ")
    natija = a + b if amal == "+" else a - b # ... va hokazo
    with open("tarix.txt", "a", encoding="utf-8") as f:
        f.write(f"{a} {amal} {b} = {natija}\n")
```

# Amaliyot: topshiriqlar

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- 1 Har bir amalni va **xato kiritishlarni** sinab ko'ring
- 2 Yangi amallar: `sqrt` (ildiz) va `%%` (foiz)
- 3 "tarix" faqat **oxirgi 5** yozuvni ko'rsatsin: `readlines()[-5:]`
- 4 `tarix.txt` ni kompyuterda topib, Bloknot bilan oching

# Ko'p uchraydigan xatolar

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- ❌ `if`, `for`, `while` oxirida `:` unutilgan → `SyntaxError`
- ❌ `else if` — Python'da `elif`
- ❌ `range(1, 10)` 10 ni bermaydi
- ❌ `"w"` rejimida muhim faylni o'chirib yuborish; qo'shish — `"a"`
- ❌ `encoding="utf-8"` siz — g'alati belgilar

# Savol

`list(range(2, 8, 2))` qanday sonlarni beradi?

**A** 2, 4, 6, 8

**B** 2, 4, 6

**C** 2, 3, 4, 5, 6, 7

**D** 8, 6, 4, 2

Javobni tanlang — to'g'ri variant yashil bo'ladi

# Savol

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## Python kod bloklarini qanday belgilaydi?

**A** Jingalak qavslar `{ }`

**B** begin / end so'zlari

**C** Nuqta-vergul

**D** Ikki nuqta va chekinish

Javobni tanlang — to'g'ri variant yashil bo'ladi

# Savol

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Fayl mazmunini o'chirmasdan oxiriga yozish rejimi?

A "a"

B "w"

C "r"

D "x"

Javobni tanlang — to'g'ri variant yashil bo'ladi

# Savol

---

`int("abc")` qanday xato beradi?

**A** ZeroDivisionError

**B** IndentationError

**C** ValueError

**D** TypeError

Javobni tanlang — to'g'ri variant yashil bo'ladi



# Uyga vazifa

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- 1 **Parol tekshiruvchi:** 8+ belgi, raqam va katta harf ( `isdigit()` , `isupper()` )
- 2 **Kundalik:** har ishga tushganda sana va xulosa → `kundalik.txt`
- 3 **Tub sonlar 1–200** → `tub.txt` , har qatorda bittadan

## Yodda tuting

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- `if / elif / else:` — `:` va 4 bo'sh joy chekinish
- `for i in range(boshi, oxiri, qadam):` — oxiri kirmaydi
- `while shart:`, `break`, `continue`
- `try: ... except ValueError: ...` — dasturni saqlash
- `with open("f.txt", "r/w/a", encoding="utf-8") as f:`



# Savollar bormi?

Keyingi mavzu: **23. Ma'lumot tuzilmalari: list, tuple, set va dictionary**

Batafsil matn, misollar va testlar: **22-python-shart-sikl-fayl.html** darsida